VNR Vignana Jyothi Institute of Engineering

Computer Science and Engineering | Course Code: CS101

Programming for Problem Solving & Object-Oriented Principles - Laborator

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Syllabus Focus: CS101 Laboratory Practical Manual & Viva Voce Handbook

Summary: Structured laboratory workbook detailing experiment objectives, baseline toolchains, e

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Imperative Foundations & Memory Layout

" Syllabus Scope: Data representations, bitwise operators, execution models, control flow, functions, cal

Unit 2: Pointers & Dynamic Memory Management

" Syllabus Scope: Pointer arithmetic, arrays, heap allocation malloc/free/new/delete, memory leak prevent

CS101: Programming for Problem Solving & Object-Oriented Principles - Advanced Topic

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Object-Oriented Architecture

" Encapsulation, abstraction, constructor/destructor chains, operator overloading, deep vs shallow copying

Unit 4: Inheritance & Dynamic Polymorphism

" Single and multiple inheritance, virtual function tables vtable, abstract base classes, runtime type identif

Unit 5: Templates, Generic Programming & STL

" Function and class templates, Standard Template Library vector, map, set, custom iterators, exception handli

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

Lab 1: Memory Layout and Pointer Arithmetic Analysis using GDB debugger

Lab 2: Object-Oriented Banking Simulation with Polymorphic Transaction Processing

Lab 3: Custom String Class Implementation with Rule of Five and Move Semantics

Lab 4: Generic Dictionary Implementation using Hash Maps with Collision Resolution

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Bjarne Stroustrup - The C++ Programming Language Addison-Wesley 4th Ed

* Robert Lafore - Object-Oriented Programming in C++ Sams Publishing

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